```
# ============================================================
#                 START: LOAD & INSPECT DATA
# ============================================================

getwd()
data_path <- "DataSet_No_Details.csv"

df <- read.csv(data_path)

# Structure and summary
str(df)

install.packages("skimr")
library(skimr)
skim(df)


# ============================================================
#                 DATA PREPARATION
# ============================================================

library(dplyr)

cols_to_remove <- c(
  "h_index_34", "h_index_56", "hormone10_1", "hormone10_2",
  "an_index_23", "outcome", "factor_eth", "factor_h",
  "factor_pcos", "factor_prl"
)

MD_df <- df %>% select(-any_of(cols_to_remove))

factor_df <- df %>% select(
  record_id, outcome, factor_eth, factor_h,
  factor_pcos, factor_prl
)

str(MD_df)
summary(factor_df)


# ============================================================
#                 MISSING DATA ANALYSIS
# ============================================================

sum(is.na(MD_df))
colSums(is.na(MD_df))
skim(MD_df)

na_stats <- colMeans(is.na(MD_df)) * 100
na_stats

# Missing ≤ 35%
data.frame(
  Column = names(na_stats[na_stats <= 35]),
  NA_Percent = na_stats[na_stats <= 35]
)

# Missing > 35%
data.frame(
  Column = names(na_stats[na_stats > 35]),
  NA_Percent = na_stats[na_stats > 35]
)
```

```
# ============================================================
#                 VISUALIZING MISSINGNESS
# ============================================================

install.packages("visdat")
library(visdat)
vis_miss(MD_df)

install.packages("naniar")
library(naniar)
gg_miss_var(MD_df)


# ============================================================
#                 PREPARE DATA FOR MCAR TEST
# ============================================================

cols_to_remove1 <- c("hormone9", "hormone11", "hormone12", "hormone13", "hormone14")
handle_MD_df <- MD_df %>% select(-any_of(cols_to_remove1))

str(handle_MD_df)


# ============================================================
#                 LITTLE'S MCAR TEST
# ============================================================

mcar_test_result <- mcar_test(handle_MD_df)
print(mcar_test_result)


# ============================================================
#                 MULTIPLE IMPUTATION (PMM)
# ============================================================

install.packages("mice")
library(mice)

imputed_handle_MD_df <- mice(handle_MD_df, m = 5, method = "pmm", print = FALSE)

# Extract completed dataset
imputed_handle_MD_df_final <- complete(imputed_handle_MD_df, 1)


# ============================================================
#                 DENSITY PLOT (Original vs Imputed)
# ============================================================

library(ggplot2)

ggplot(handle_MD_df, aes(x = hormone10_generated, fill = "Original")) +
  geom_density(alpha = 0.5) +
  geom_density(
    data = imputed_handle_MD_df_final,
    aes(x = hormone10_generated, fill = "Imputed"),
    alpha = 0.5
  ) +
  labs(title = "Density Plot of hormone10_generated: Original vs Imputed") +
  scale_x_continuous(limits = c(0, 2))


# ============================================================
#                 OUTLIER DETECTION (LIPID VARIABLES)
# ============================================================
```

```
# Select lipid variables
lipid_df <- imputed_handle_MD_df_final %>%
  select(lipids1, lipids2, lipids3, lipids4, lipids5)

# Outlier detection function
detect_outlier_direction <- function(x) {
  Q1 <- quantile(x, 0.25, na.rm = TRUE)
  Q3 <- quantile(x, 0.75, na.rm = TRUE)
  IQR_val <- Q3 - Q1

  lower_bound <- Q1 - 1.5 * IQR_val
  upper_bound <- Q3 + 1.5 * IQR_val

  ifelse(x < lower_bound, "Lower Outlier",
         ifelse(x > upper_bound, "Upper Outlier", "Normal"))
}

# Apply outlier detection
outlier_df <- lipid_df %>%
  mutate(across(everything(), detect_outlier_direction))

# Convert to long format
outlier_long <- outlier_df %>%
  pivot_longer(cols = everything(),
               names_to = "Variable",
               values_to = "Outlier_Type")


# ==============================================================
#                 GRAPH 1: BOXPLOT (LIPID VARIABLES)
# ==============================================================

lipid_df %>%
  pivot_longer(everything(), names_to = "Variable", values_to = "Value") %>%
  ggplot(aes(x = Variable, y = Value)) +
  geom_boxplot(fill = "lightblue", alpha = 0.7) +
  labs(title = "Boxplots for Lipid Variables",
       y = "Value") +
  theme_minimal()


# ==============================================================
#                 GRAPH 2: OUTLIER DIRECTION BAR CHART
# ==============================================================

ggplot(outlier_long, aes(x = Variable, fill = Outlier_Type)) +
  geom_bar(position = "dodge") +
  labs(title = "Outlier Direction by Lipid Variable",
       y = "Count",
       x = "Lipid Variables") +
  theme_minimal()


# ==============================================================
#                 GRAPH 3: BOXPLOTS FOR ALL NUMERIC VARIABLES
# ==============================================================

imputed_handle_MD_df_final %>%
  select(where(is.numeric)) %>%
  pivot_longer(everything(),
               names_to = "Variable",
               values_to = "Value") %>%
  ggplot(aes(x = Variable, y = Value)) +
  geom_boxplot(fill = "skyblue", alpha = 0.7) +
```

```
    facet_wrap(~ Variable, scales = "free") +
    labs(title = "Boxplots for All Numeric Variables",
         x = "",
         y = "Value") +
    theme_minimal() +
    theme(axis.text.x = element_blank(),
          axis.ticks.x = element_blank())

  library(dplyr)
  library(tidyr)
  library(ggplot2)

  # Ensure dataset exists
  imputed_handle_MD_df <- mice(handle_MD_df, m = 5, method = "pmm", print = FALSE)
  imputed_handle_MD_df_final <- complete(imputed_handle_MD_df, 1)

  # Build the graph
  imputed_handle_MD_df_final %>%
    select(where(is.numeric)) %>%
    pivot_longer(everything(),
                 names_to = "Variable",
                 values_to = "Value") %>%
    ggplot(aes(x = Variable, y = Value)) +
    geom_boxplot(fill = "skyblue", alpha = 0.7) +
    facet_wrap(~ Variable, scales = "free") +
    theme_minimal() +
    theme(axis.text.x = element_blank(),
          axis.ticks.x = element_blank()) +
    labs(title = "Boxplots for All Numeric Variables",
         y = "Value")
```